

Arpan Pathak

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📌 SUMMARY

Systems Software Engineer with **8 years** architecting high-performance, low-latency systems at **Amazon**, **Microsoft**, and **Oracle**: database internals, kernel-level networking, distributed systems, and real-time ML inference. Deep expertise in **Rust**, **C++**, **Go**, and **C**: lock-free concurrency, memory safety, eBPF/Cilium, and GPU-accelerated computing. Shipped measurable wins: ~ **40% lower P99 latency**, **2x throughput**, **45,000 ops/s** throughput gains, **5,000 QPS** inference serving, and **\$5.4M/month** revenue impact.

📌 PERSONAL PROJECTS

- 📁 **Driving-CivicSense (Edge AI Vision)** | github.com/arpanpathak/driving-civicsense-vision-model | 2026
- On-device AI perception system built in **Rust** for intersection discipline and lane awareness, running entirely on-device (< **50ms inference latency**, no cloud dependency) for aftermarket dashcam and AR-glasses form factors.
 - Fine-tunes **YOLOv8/v11** and custom **CNN** architectures; applies model pruning and quantization to fit tight memory and power budgets on low-power ARM single-board computers (Raspberry Pi Zero, Raspberry Pi 5, and other low-power ARM processor boards).
 - Trains on cloud GPU infrastructure with synthetic data augmentation; planning to release a public driving-behavior dataset on **Hugging Face**.
- 📖 **Companion Book: Seeing Machines: Deep Learning & Computer Vision from Python to Bare Metal** | arpanpathak.github.io/seeing-machines-book
- 📖 **Physics & Mathematics of Game Development in Rust (Book)** | arpanpathak.github.io/bevy-physics-book | 2026
- 📖 **CUDA Kernels: GPU & Parallel Programming from First Principles (Book)** | arpanpathak.github.io/gpu-parallel-book | github.com/arpanpathak/gpu-parallel-book | 2026

📌 EXPERIENCE

Software Developer II, Microsoft, Redmond, WA

Jun 2025 - Aug 2026

- C++/C# to Rust Migration (Memory Safety)**: Spearheaded migration of the Information Protection service from **unsafe C++** and **C#** to **Rust**, moving memory safety to compile time: eliminated use-after-free, buffer-overflow, and null-safety bugs in the migrated path. Delivered ~ **40% lower P99 latency**, **2x throughput**, **3x lower memory** per request, and **6-figure annual cost savings**.
 - Zero-Trust Kernel Networking (eBPF/Cilium)**: Architected zero-trust sovereign cloud networking with **eBPF + Cilium** network policies enforcing **data residency** and **SecNumCloud** compliance; authored **Kubernetes operators (Kubebuilder)** to reconcile policy state across AKS clusters in France and Germany.
 - AI-Powered Policy Engine**: Built an AI engine that auto-generates Cilium policies and scans for misconfigurations: manual review cut from **4 hours** to **near-real-time**, zero policy drift across sovereign regions.
 - CI/CD & Observability**: Containerized microservices with blue-green deployments, **Prometheus/Azure Monitor**, and automated validation for reliable multi-region releases.
- 🔗 **TechStack**: **Rust, C++, eBPF, Cilium, Kubernetes (AKS), Kubebuilder, Prometheus, Linux**

Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA

Oct 2024 - Mar 2025

- High-Performance Database Internals**: Architected low-latency internals for a proprietary database engine, optimizing **lock-free data structures** to cut contention in high-concurrency read-write paths: **45,000 ops/s** transaction throughput gain.
 - Infrastructure-as-Code SDK**: Engineered a **Terraform provider** (developer-facing SDK) for Oracle Autonomous Database, replacing legacy control-plane tooling with **\$1.2M/month** projected savings.
 - Security & Key Management**: Integrated OCI Security Vault across control-plane services for secret management and automated key rotation.
- 🔗 **TechStack**: **Java, Go, Terraform, Oracle Cloud Infrastructure, Autonomous Database**

Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India

Mar 2021 - Oct 2024

- **Real-Time ML Inference Engine:** Architected **XGBoost + Deep Learning ensemble** ranking for ads & deals across FreeVee, Twitch, Prime Video, and Amazon Retail: millions of events at **100ms inference latency**, driving **5% CTR lift**, **30%** more deal impressions, and **\$5.4M/month revenue growth**.
- **BERT Inference Service:** Delivered **BERT-based** NLP inference handling **5,000 QPS** at peak, cutting support-ticket triage SLA from **7 days to 2 hours**.
- **Distributed Systems & SDK Design:** Designed uniqueness-constraint indexing and an **SDK** for a multi-tenant NoSQL store using **two-phase commit**, achieving **serializable isolation** at **3K writes/second**: a developer-facing library used by multiple publisher teams.
- **Event Bus SDK (Kotlin, CSP):** Built an event bus SDK using **Kotlin structured concurrency** and the **CSP (communicating sequential processes)** paradigm to deliver programmatic guaranteed ad platform deals in real-time with **< 50ms end-to-end latency**.
- **Big Data Platform:** Built big data infrastructure & pipelines for the Amazon WorkEvents data platform stream team: **1-petabyte** data warehouse & data lake serving **1M+ daily BI intelligence queries**. Wrote business logic in **AWS Glue (serverless Apache Spark)** to process & transform data for **financial reporting and pay computation**, with a **DSL execution engine** for OLTP and BI use cases.

⟨ TechStack: **Kotlin, Java, Python, PyTorch, XGBoost, Spark, Kafka, DynamoDB, Redshift, AWS**

Software Development Engineer, Razorpay, Bengaluru, India

Aug 2020 - Feb 2021

- Built UPI Payment Gateway Aggregator in **Go** + Protobuf processing **1M+ hourly transactions** with **99.99% uptime** at **150ms P99** under **50,000 TPM** peak load.

⟨ TechStack: **Go, Redis, PostgreSQL, Protobuf, Kafka**

Software Engineer, Mindfire Solutions, Bhubaneswar, India

Aug 2018 - Feb 2020

- Delivered full-stack workflow tools and an **AR-based product search microservice** serving **10,000+ daily queries**; cut dashboard latency from **5 min to 1 min**.

⟨ TechStack: **Java, Spring Boot, Angular, MySQL, MongoDB**

SKILLS

⟨ **Languages:** Rust, C++, Go, C, Python, Java, Kotlin, TypeScript, Shell Script

⚙️ **Systems Programming & Performance:** Lock-Free Data Structures, Concurrency, Memory Safety, Memory Profiling, Cache Optimization, Performance Engineering, Low-Latency Systems, OS Internals, Computer Architecture

🌐 **Networking & Protocols:** TCP/IP, UDP, TLS, mTLS, DNS, HTTP/2, HTTP/3, QUIC, WebSockets, RDMA, InfiniBand, DPDK, VXLAN, GRE, Socket Programming, Packet Processing, Network Performance Optimization

🔒 **Network & Cloud Security:** eBPF, XDP, Cilium, Kernel-Level Networking, Packet Processing, VPC, Subnets, Routing, NAT, VPN, Peering, Load Balancing, Security Groups / Firewalls

📦 **Distributed Systems & Streaming:** Distributed Systems (consensus, replication, sharding), Apache Kafka, Apache Flink, Apache Spark, Spark Streaming, AWS Kinesis, Real-Time Data Pipelines, Change Data Capture (CDC), Microservices

🔗 **APIs & Serialization:** gRPC, REST API Design, Protobuf

⚡ **ML Inference & AI:** PyTorch, TensorFlow, XGBoost, Transformer Models (BERT), LLMs, ONNX Runtime, On-Device/Edge Inference, Model Pruning & Quantization, Computer Vision (YOLOv8/v11), ANN, CNN, NLP, Deep Learning, Recommendation Systems, MLOps

⚙️ **GPU & Parallel Computing:** CUDA, CUDA-Oxide (Rust-to-CUDA), PTX, SIMT, GPU Kernel Programming, Parallel Computing, GPU-Accelerated Computing

☁️ **Cloud & Infra:** AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS), Google Cloud, Kubernetes (EKS, AKS), Docker, Terraform, Kubebuilder, Istio, Kubernetes Gateway API

📊 **Observability:** Prometheus, Grafana, Azure Monitor, AWS CloudWatch

📄 **Data & Storage:** DynamoDB, PostgreSQL, Redis, MongoDB, MySQL, Amazon Redshift, Amazon S3

🔑 **Core CS:** Data Structures & Algorithms, System Design, Operating Systems, Linux, Computer Architecture

EDUCATION

Bachelor of Technology in Computer Science & Engineering

RCC Institute of Information Technology, Kolkata, India | 2018